Solution

Let 'x' be the number of students who participate the tour.

Then $35 \leq x \leq 50$ and x is a positive integer.

Formula : ***Total cost = (cost per student) × (number of students)***

(i) If the number of students are between 35 and 45, then

the cost per student is ₹ 200

$\therefore$ The total cost is $y = 200x$.

(ii) If the number of students are between 46 and 50, then

the cost per student is ₹ $\left\{200 - \frac{1}{5}(x - 45)\right\}$

$= 209 - \frac{x}{5}$

$\therefore$ The total cost is $y = \left(209 - \frac{x}{5}\right)x = 209x - \frac{x^2}{5}$

$$\therefore y = \begin{cases} 200x & ; \ 35 \leq x \leq 45 \\ 209x - \frac{x^2}{5} & ; \ 46 \leq x \leq 50 \end{cases}$$ and

the domain is {35, 36, 37, ..., 50}.

Example 5.10

Draw the graph of the function

$f(x) = |x|$

Solution

$y = f(x) = |x|$

$$= \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Choose suitable values for x and determine y.

Thus we get the following table.

x	−2	−1	0	1	2
y	2	1	0	1	2

Table : 5.1

Plot the points (−2, 2), (−1, 1), (0, 0), (1, 1), (2, 2) and draw a smooth curve.

The graph is as shown in the figure

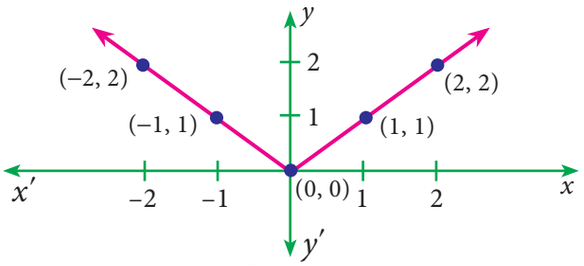


Fig. 5.4

Example 5.11

Draw the graph of the function $f(x) = x^2 - 5$

Solution

Let $y = f(x) = x^2 - 5$

Choose suitable values for x and determine y.

Thus we get the following table.

x	−3	−2	−1	0	1	2	3
y	4	−1	−4	−5	−4	−1	4

Table : 5.2

Plot the points (−3, 4), (−2, −1), (−1, −4), (0, −5), (1, −4), (2, −1), (3, 4) and draw a smooth curve.

The graph is as shown in the figure

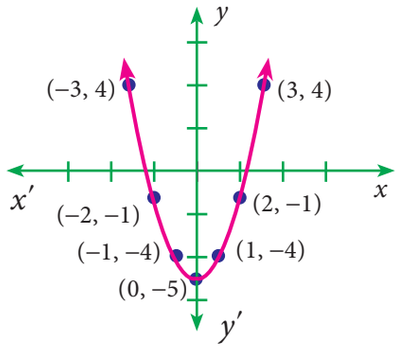


Fig 5.5

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